Lecture 4

Monday, 23 January 2023 8:32 AM

Qualifying Exam:

+ Written Report : Summary — Due: Sept. 15
  Intro
  Research Objective
  Literature Review & Theory
  Research plan
  Materials & Methods
  Preliminary Results
  Plan for next 12 months

+ Oral Report : 20 min (Uninterrupted) — Due: Oct. break!
  No back-up slide
  30 min of questions → general engineering knowledge
  → Details about methods

+ Research hypothesis (Problem definition)
  → Do not be vague
  → Define the rationale (why do you think that (your expectation) happens?)
  → Should be testable
  → justification of why you think you could do / falsify

Become skeptical → Why? How?

Hypothesis vs. problem definition

+ Evaluation : Current state-of-the art in field
  1) Defend underlying basis (why are you doing what you are doing)

  2) Impact of the proposed work : Why is this imp. to other ChemE's?
     Why should people care? → Read review papers to gain better understanding
     How does your work fit into

  3) Preliminary Results & Mastery of key Research Techniques :
     → point out the limitations of your approach/idea → and what you are going to do to deal w/ it
     → design of experiment
     → Independence
     → why & how (the machine works & what the results mean?)
     → does the data prove or disprove the initially-proposed hypothesis

4) Potential of the project for Tangible Product generation
- → Trajectory (Research plan)
- → Can change direction / report failure
- → Careful analysis of why things went wrong

5) Communication Ability:
- → Explain to other chemE's
- → Graph → clear, easily readable
- → Animations
- → enthusiastic voice

3 faculty; Grading: 2 "not passing" on 2 same fields → fail

Written: 95% pass rate

Oral: 75% " for first time

Full-page feedback, retake w/ 3 new faculty